EXPERIMENT TO UNDERSTAND LOGISTIC RGRESSION FOR A DATASET

Program:

```
 import numpy as np
import pandas as pd

df = pd.read_csv('Social_Network_Ads_modified.csv')
df
df.head()
features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values

Features
Label
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=37
model = LogisticRegression()

model.fit(x_train, y_train)
) train_score = model.score(x_train, y_train)
test_score = model.score(x_test, y_test)

print("Train Accuracy:", train_score)
```

print("Test Accuracy:", test_score)

from sklearn.metrics import classification_report

print(classification_report(label, model.predict(features)))

OUTPUT:

	User ID	Gender	Age	EstimatedSalary	Purchased	
0	17000001	Male	41	74570	1	
1	17000002	Female	28	19015	0	
2	17000003	Male	25	79813	0	
3	17000004	Male	53	45712	0	
4	17000005	Male	55	118497	1	
...	...	...	...	...	...	...
395	17000396	Female	22	57986	1	
396	17000397	Female	42	103595	1	
397	17000398	Female	19	119358	1	
398	17000399	Female	27	82929	1	
399	17000400	Male	47	91506	1	

400 rows × 5 columns

	User ID	Gender	Age	EstimatedSalary	Purchased	
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2	17000003	Male	25	79813	0	
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x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=37
)
Train Accuracy: 0.7375
Test Accuracy: 0.675

precision    recall  f1-score   support

           0       0.63      0.29      0.40       125
           1       0.74      0.92      0.82       275

    accuracy                           0.72       400
   macro avg       0.69      0.61      0.61       400
weighted avg       0.71      0.72      0.69       400
```